FlightOps

Project Proposal Report

OVERVIEW

1. Project background and description

Background:

Bad weather frequently delays flights, costing time and money. Traditional systems only respond after delays occur, making proactive management difficult.

Description:

FlightOps is a real-time flight tracking system. A custom ingestion script continuously fetches live flight coordinates from the **OpenSky Network API** and combines them with weather data. The platform analyzes these conditions to predict flight delay risks and serves the results through a simple REST API.

2. Project Objectives

The main goal of FlightOps is to build a scalable, cloud-native infrastructure that automates the entire software lifecycle. The key objectives are:

Automate Data Collection: Fetch live flight data from **OpenSky API** automatically using a dedicated script.

Predict Flight Delays: Analyze flight telemetry alongside weather conditions to forecast dlay risks.

Send Automated Alerts: Instantly notify the team/system when high delay risks or system errors occur.

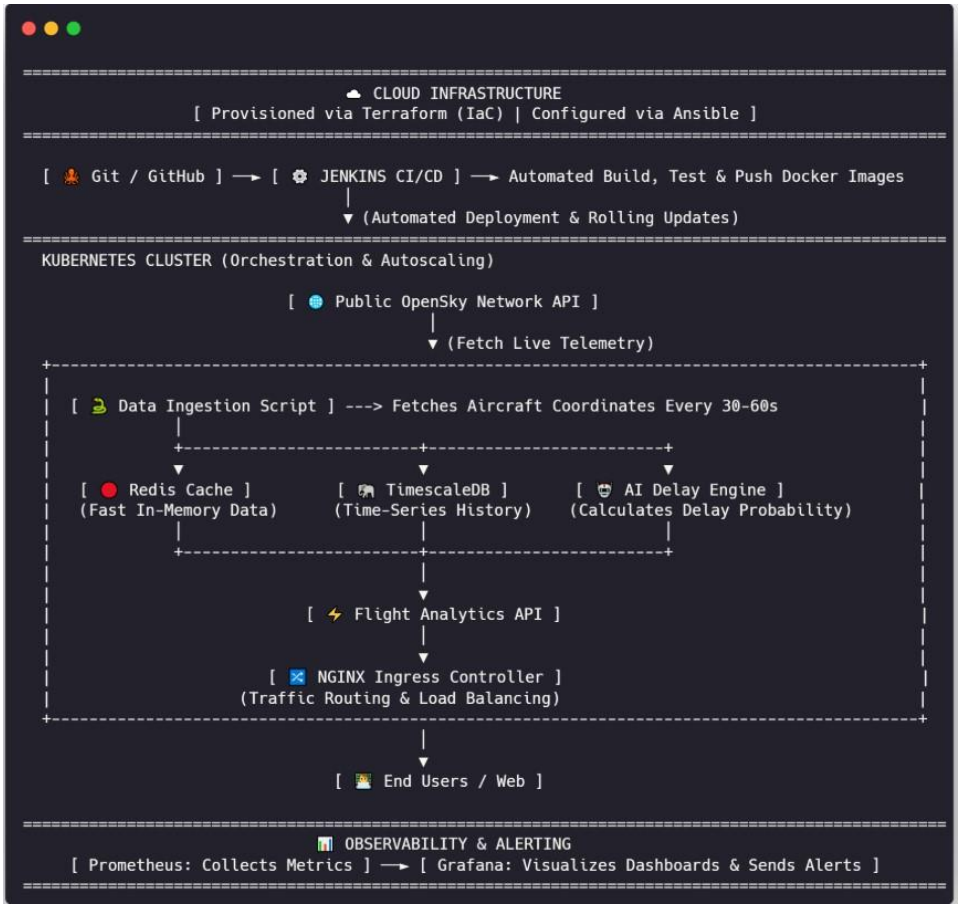
Implement DevOps Best Practices: Use containerization, Infrastructure as Code (IaC), and automated deployment.

Ensure High Availability & Monitoring: Maintain zero downtime and continuously track system metrics and logs.

Tools & Technologies

Data Ingestion Script (Python/Go):	**Automatically fetches live aircraft data from OpenSky API.**
Git & GitHub:	Manages code versioning and collaboration among the team
Docker:	Package the ingestion script and microservices into lightweight containers
Kubernetes (K8s):	Manages container deployment, scaling, and high availability
Terraform:	Automates provisioning cloud infrastructure (IaC)
Cloud (AWS)	Hosts the servers, network, and Kubernetes cluster
Ansible:	Automates server setup, package installation, and initial configuration
Jenkins:	Runs the CI/CD pipeline to automate testing, building, and deployment
NGINX / Ingress Controller:	Manages web server configuration and routes traffic to the services.
Application Deployment	Applies zero-downtime deployment strategies (Rolling Updates)
Monitoring (Prometheus & Grafana):	Tracks system metrics and sends automated alerts for system issues.
AI	Processes live flight data alongside historical weather patterns to predict flight delay risks and calculate probability scores.

(SYSTEM ARCHITECTURE DIAGRAM)



TEAM :

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